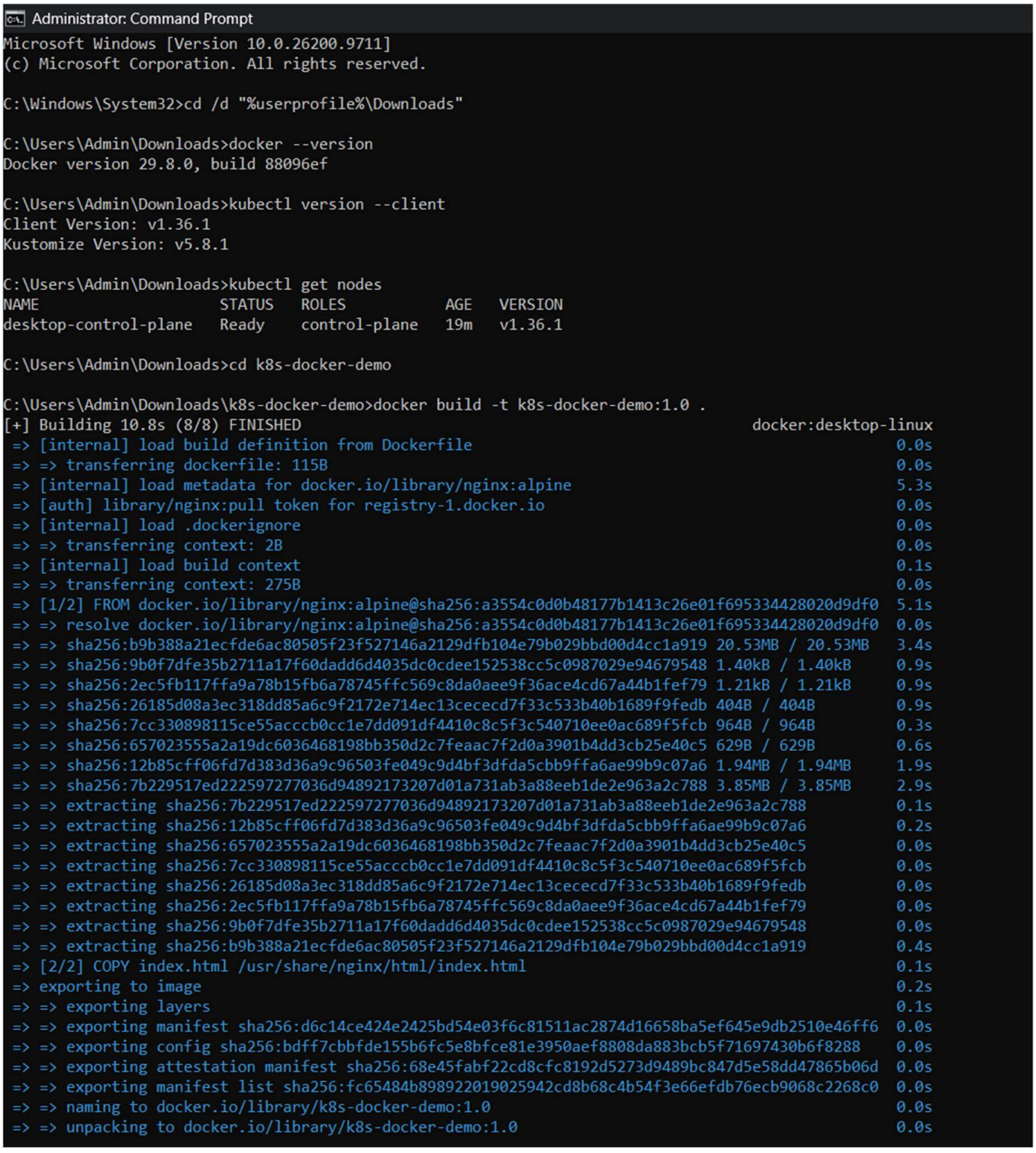
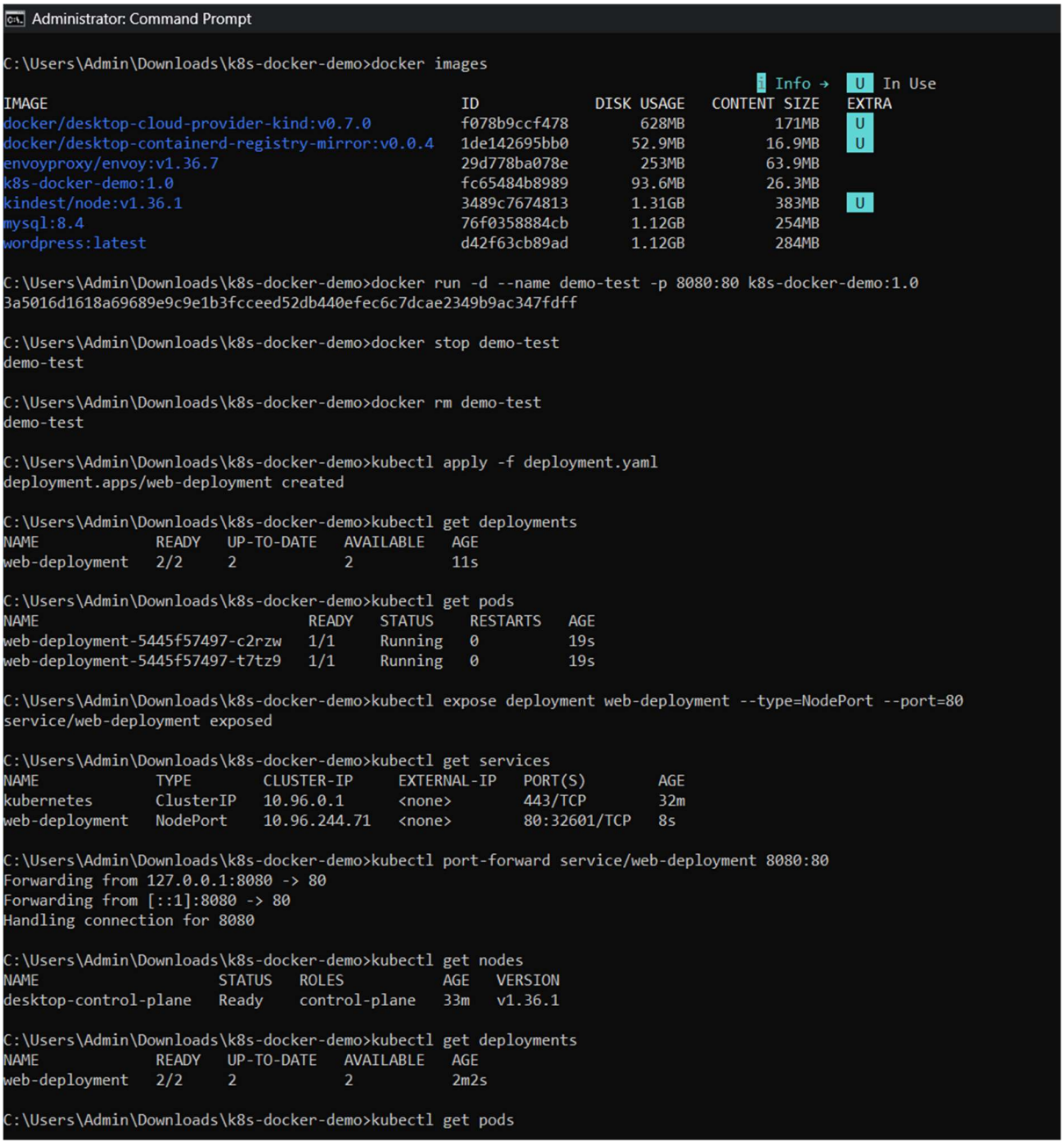
Experiment No. 10

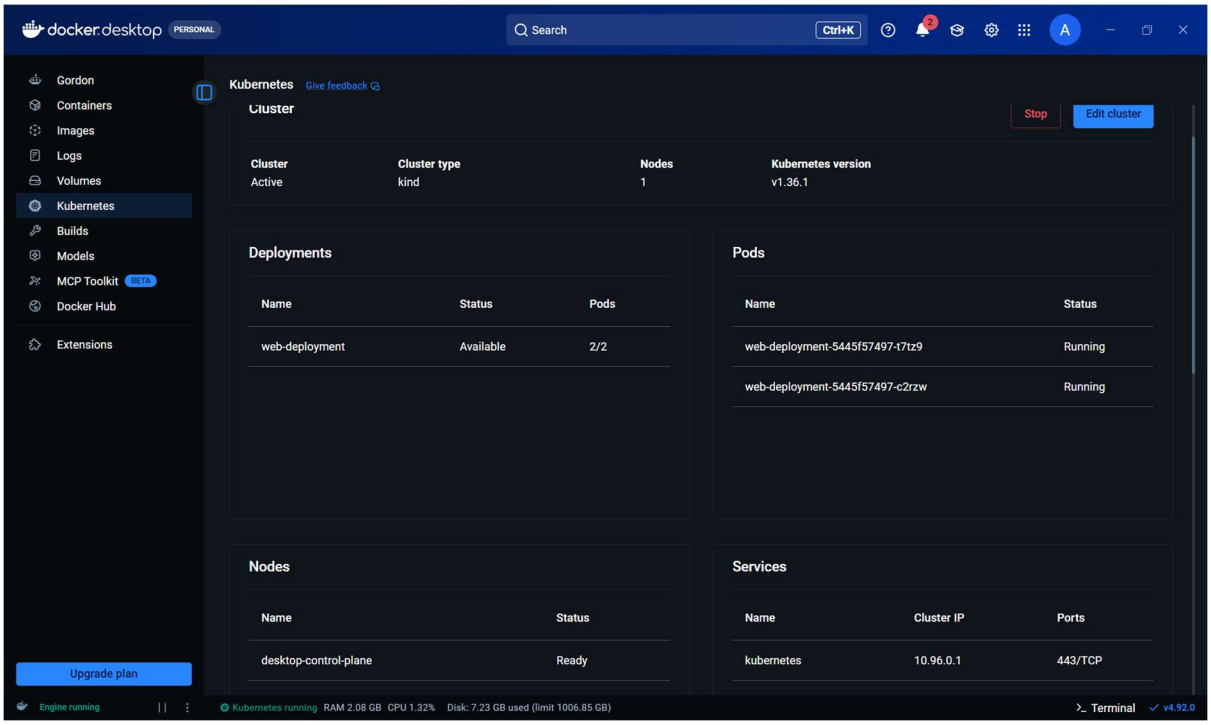
Kubernetes and Docker Integrate

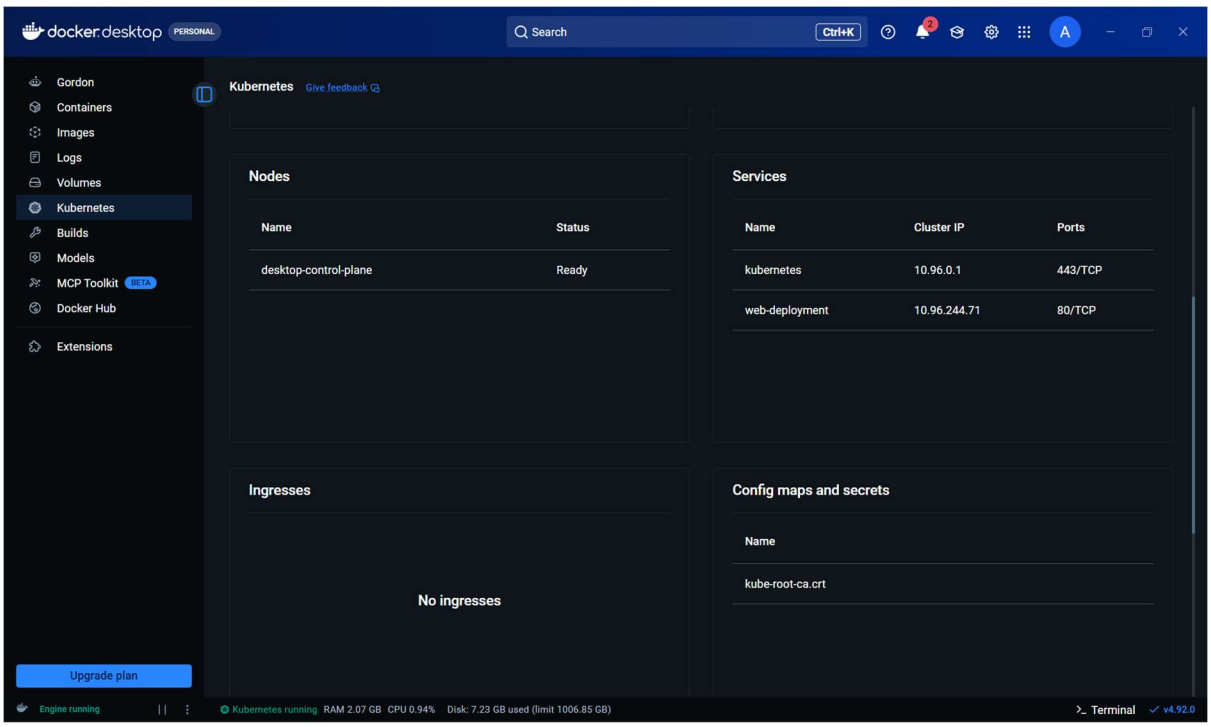
NAME : Aayush Patil PRN : 2324000488

SUB : DevOps lab ROLL NO : 31 Batch: A-1




Administrator: Command Prompt
C:\Users\Admin\Downloads\k8s-docker-demo>docker images
i Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
docker/desktop-cloud-provider-kind:v0.7.0 f078b9ccf478 628MB 171MB U
docker/desktop-containerd-registry-mirror:v0.0.4 1de142695bb0 52.9MB 16.9MB U
envoyproxy/envoy:v1.36.7 29d778ba078e 253MB 63.9MB
k8s-docker-demo:1.0 fc65484b8989 93.6MB 26.3MB
kindest/node:v1.36.1 3489c7674813 1.31GB 383MB U
mysql:8.4 76f0358884cb 1.12GB 254MB
wordpress:latest d42f63cb89ad 1.12GB 284MB
C:\Users\Admin\Downloads\k8s-docker-demo>docker run -d --name demo-test -p 8080:80 k8s-docker-demo:1.0
3a5016d1618a69689e9c9e1b3fcceed52db440efec6c7dcae2349b9ac347fdff
C:\Users\Admin\Downloads\k8s-docker-demo>docker stop demo-test
demo-test
C:\Users\Admin\Downloads\k8s-docker-demo>docker rm demo-test
demo-test
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl apply -f deployment.yaml
deployment.apps/web-deployment created
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
web-deployment 2/2 2 2 11s
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl get pods
NAME READY STATUS RESTARTS AGE
web-deployment-5445f57497-c2rzw 1/1 Running 0 19s
web-deployment-5445f57497-t7tz9 1/1 Running 0 19s
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl expose deployment web-deployment --type=NodePort --port=80
service/web-deployment exposed
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 32m
web-deployment NodePort 10.96.244.71 <none> 80:32601/TCP 8s
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl port-forward service/web-deployment 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl get nodes
NAME STATUS ROLES AGE VERSION
desktop-control-plane Ready control-plane 33m v1.36.1
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
web-deployment 2/2 2 2 2m2s
C:\Users\Admin\Downloads\k8s-docker-demo>kubectl get pods


docker desktop
PERSONAL
Search
Ctrl+K
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Upgrade plan
Engine running
Kubernetes
Give feedback
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kind
Nodes
1
Kubernetes version
v1.36.1
Deployments
Name
Status
Pods
web-deployment
Available
2/2
Pods
Name
Status
web-deployment-5445f57497-t7tz9
Running
web-deployment-5445f57497-c2rzw
Running
Nodes
Name
Status
desktop-control-plane
Ready
Services
Name
Cluster IP
Ports
kubernetes
10.96.0.1
443/TCP
Kubernetes running
RAM 2.08 GB
CPU 1.32%
Disk: 7.23 GB used (limit 1006.85 GB)
Terminal
v4.92.0


docker desktop
PERSONAL
Search
Ctrl+K
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Upgrade plan
Engine running
Kubernetes
Give feedback
Nodes
Name
Status
desktop-control-plane
Ready
Services
Name
Cluster IP
Ports
kubernetes
10.96.0.1
443/TCP
web-deployment
10.96.244.71
80/TCP
Ingresses
No ingresses
Config maps and secrets
Name
kube-root-ca.crt
Kubernetes running
RAM 2.07 GB
CPU 0.94%
Disk: 7.23 GB used (limit 1006.85 GB)
Terminal
v4.92.0



Experiment No. 10

Docker and Kubernetes Integration

Application deployed successfully using Kubernetes!

